



BITS Pilani

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CE F231

Fluid Mechanics

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Lecture 14: Fluid Kinematics

Velocity



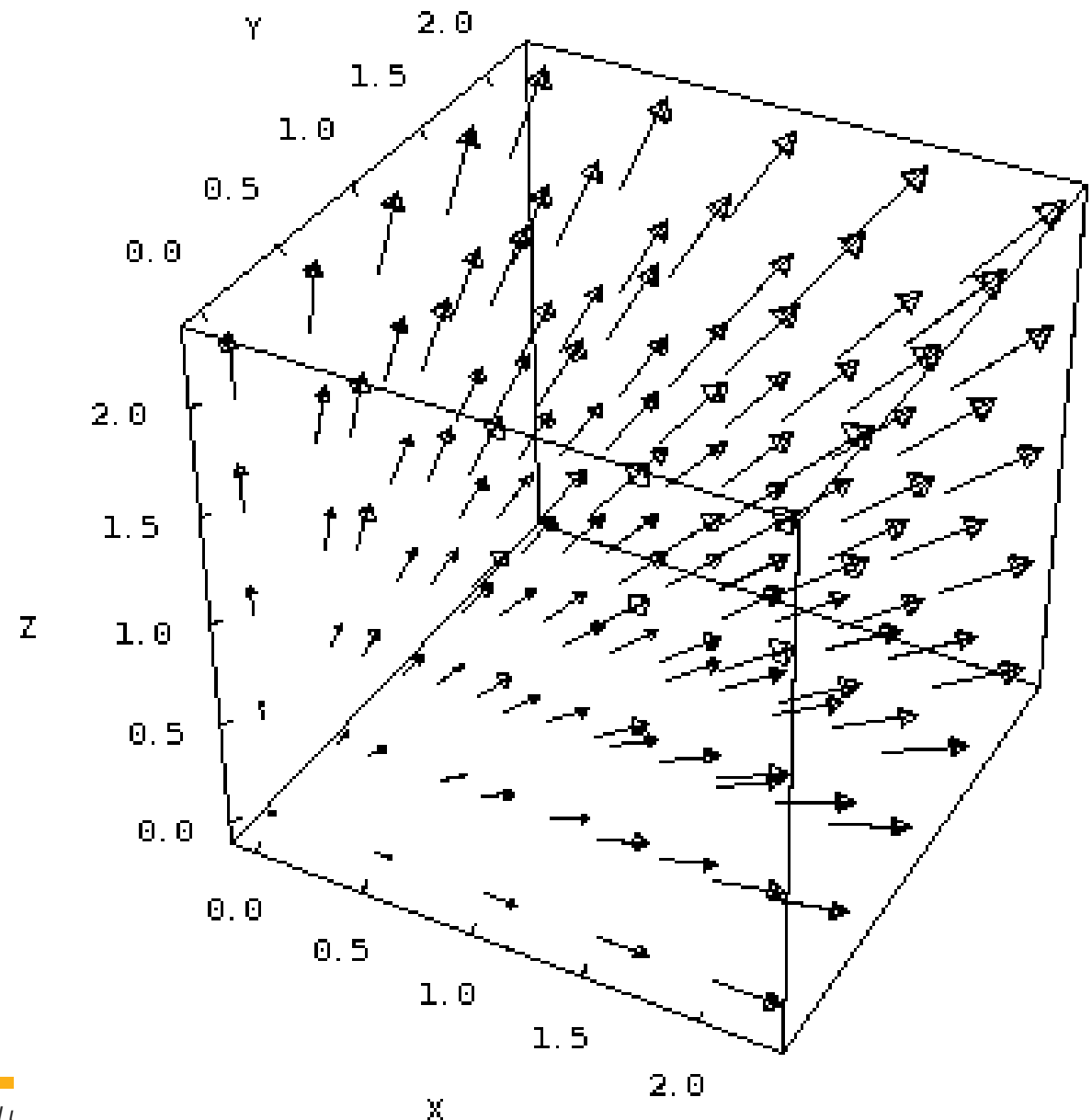
Velocity Field

$$u = f_1(x, y, z, t)$$

$$v = f_2(x, y, z, t)$$

$$w = f_3(x, y, z, t)$$

$$\mathbf{V} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$$



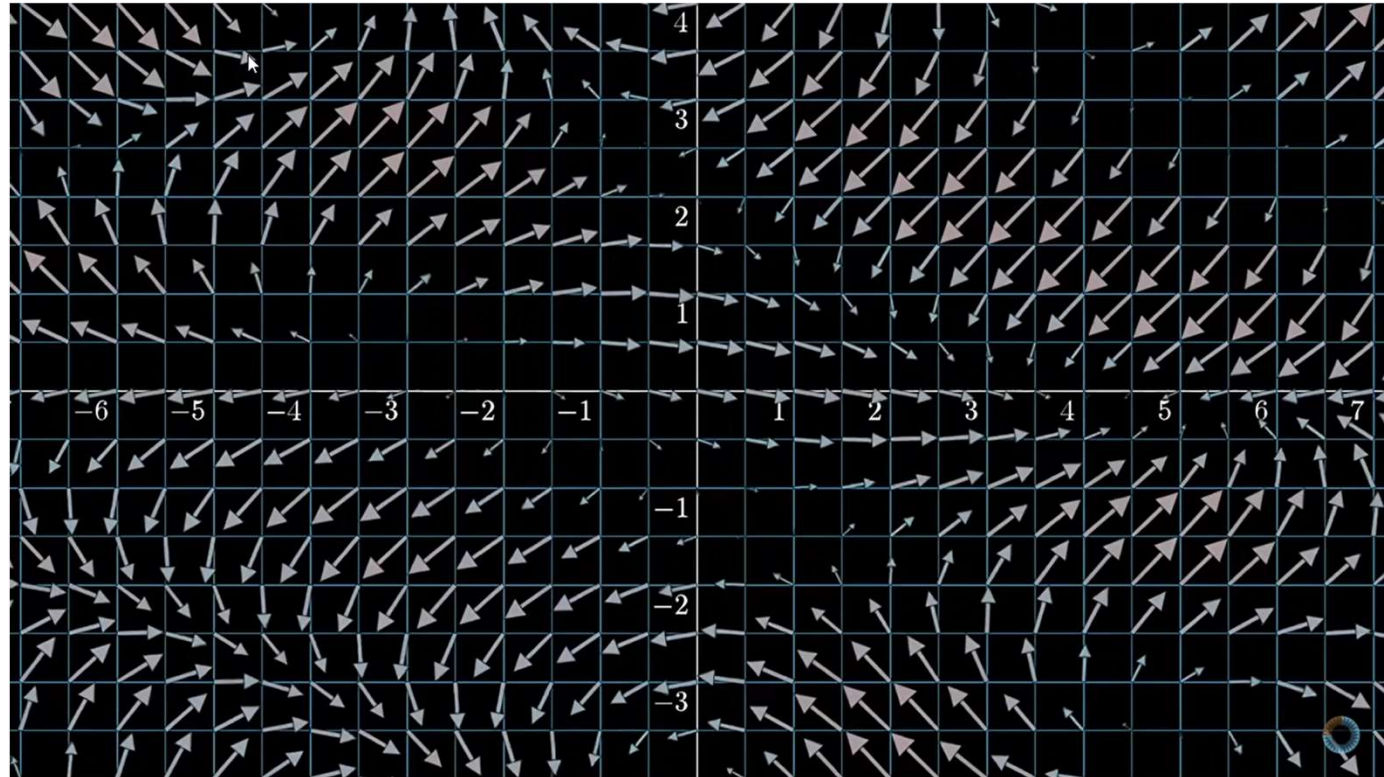
Velocity Field

$$u = f_1(x, y, z, t)$$

$$v = f_2(x, y, z, t)$$

$$w = f_3(x, y, z, t)$$

$$\mathbf{V} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$$



Velocity Field



Velocity Field (Experimental – PIV)



Velocity Field



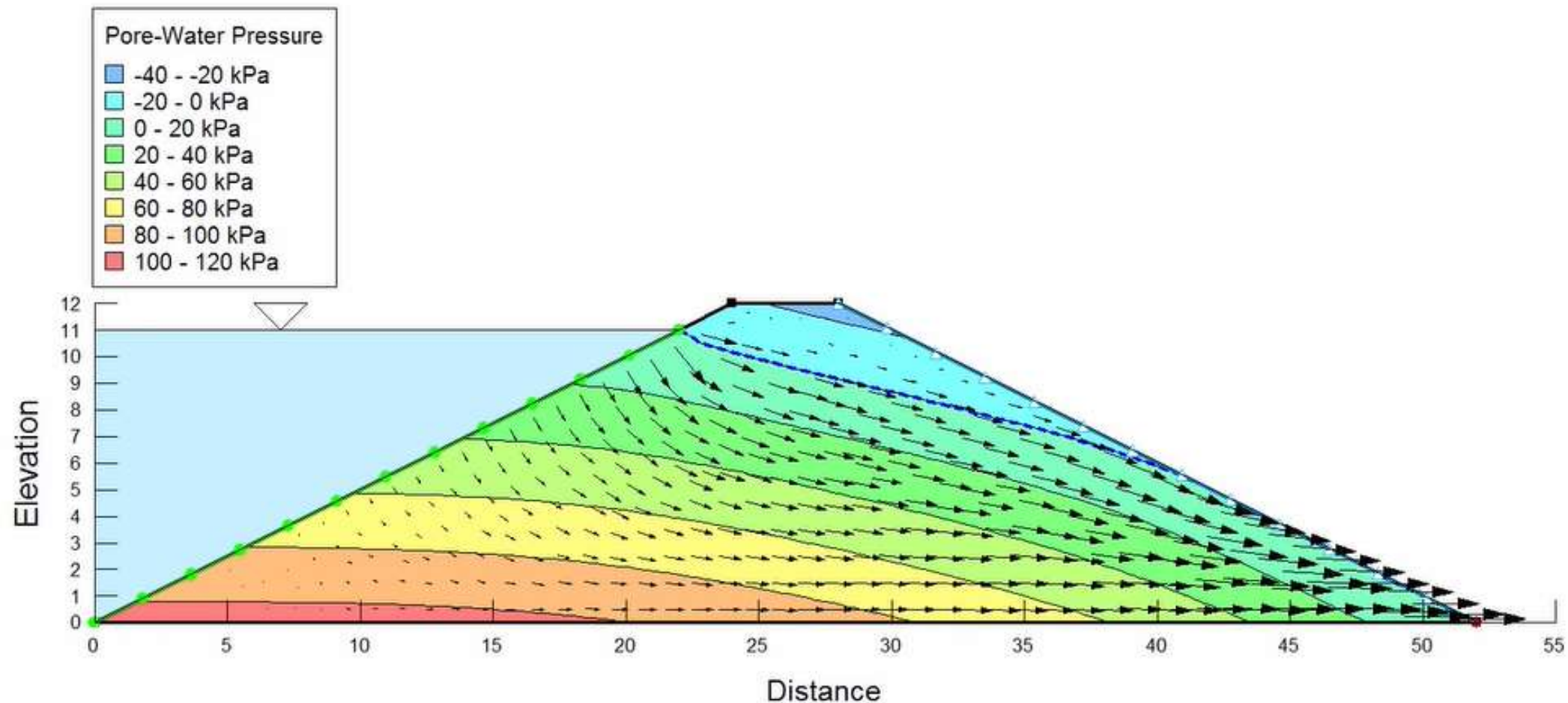
Velocity Field (Experimental – PIV)



Velocity Field



Velocity Field (Numerical Simulation – Geostudio SEEP/W)



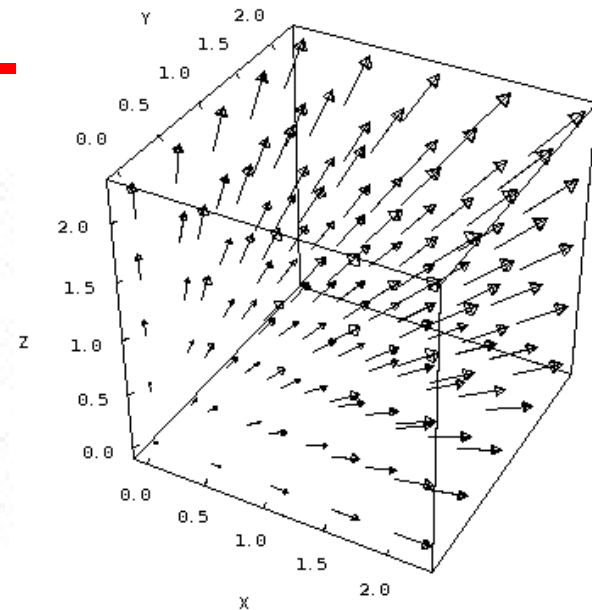
Velocity and Acceleration

Velocity

x-component of velocity $u = f_1(x, y, z, t)$

y-component of velocity $v = f_2(x, y, z, t)$

z-component of velocity $w = f_3(x, y, z, t)$



Resultant velocity, $V = ui + vj + wk$

Acceleration

$$a_x = \frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} + \frac{\partial u}{\partial t}$$

Velocity and Acceleration

Acceleration

$$a_x = \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

$$a_y = \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

Velocity and Acceleration

Local/Temporal acceleration

Local acceleration is rate of change in velocity with respect to time at a given point in flow field

$$a_x = \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

$$a_y = \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

Velocity and Acceleration

Convective acceleration

Convective acceleration is rate of change in velocity due to change in position of fluid particles in a flow field

$$a_x = \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

$$a_y = \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

Velocity and Acceleration

- Total acceleration = Convective acceleration + Local acceleration

Total	Convective	Local
$a_x = \frac{du}{dt}$	$= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$	$+ \frac{\partial u}{\partial t}$
$a_y = \frac{dv}{dt}$	$= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$	$+ \frac{\partial v}{\partial t}$
$a_z = \frac{dw}{dt}$	$= u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z}$	$+ \frac{\partial w}{\partial t}$

Question

- Acceleration for Steady Flow?
- Steady flow: Fluid characteristics (e.g., velocity) at any point in fluid flow do not change with time

Convective

Local

$$\begin{aligned}
 a_x &= \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \\
 a_y &= \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \\
 a_z &= \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}
 \end{aligned}$$

Question

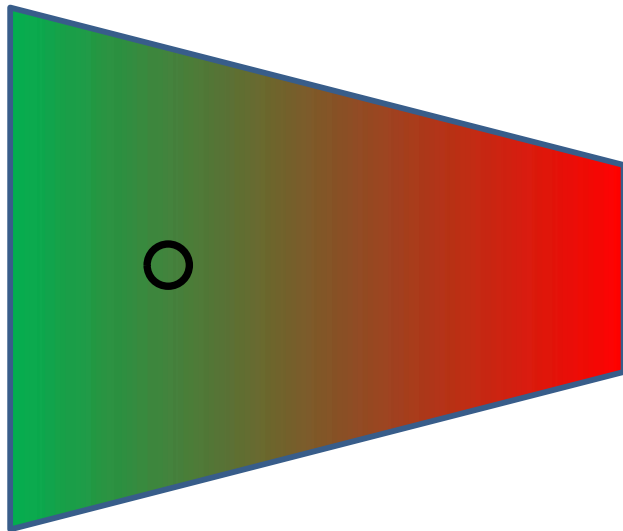
➤ Steady Flow

- Steady Flow: Fluid characteristics at any point in fluid flow do not change with time

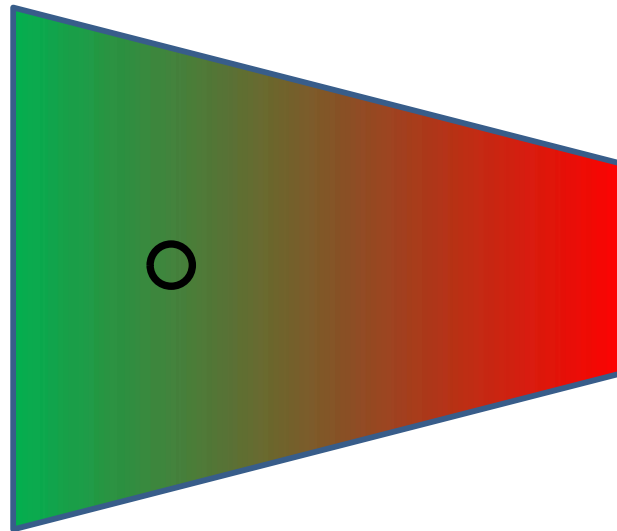
$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial \rho}{\partial t}\right)_{x_0, y_0, z_0} = 0$$

where (x_0, y_0, z_0) is a fixed point in fluid field.

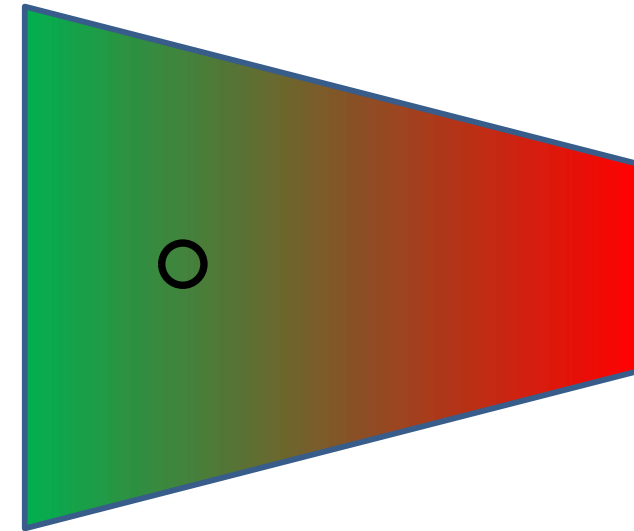
10 m/s 20 m/s 30 m/s



At time $t = t_1$



At time $t = t_2$



At time $t = t_3$

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Question

- Acceleration for Uniform Flow?
- Uniform Flow: Fluid characteristics at any time do not change with space

Convective
Local

$$\begin{aligned}
 a_x &= \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \\
 a_y &= \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \\
 a_z &= \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}
 \end{aligned}$$

Question

➤ Uniform Flow

- Uniform Flow: Fluid characteristics at any time do not change with space

$$\left(\frac{\partial V}{\partial s} \right)_{t = \text{constant}} = 0$$

where ∂V = Change of velocity
 ∂s = Length of flow in the direction S .

10 m/s 20 m/s 30 m/s



At time $t = t_1$



At time $t = t_2$



At time $t = t_3$

Question

- Acceleration for Steady and Uniform Flow?

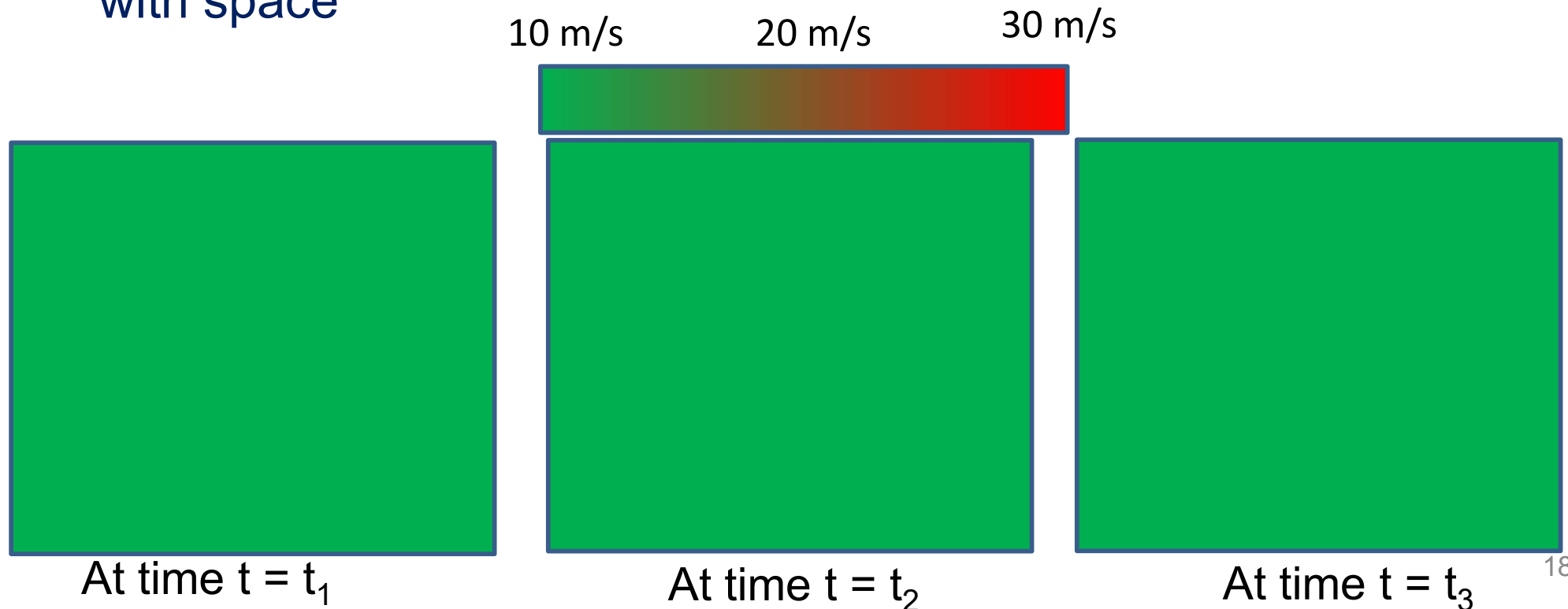
Convective
Local

$$\begin{aligned}
 a_x &= \frac{du}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \\
 a_y &= \frac{dv}{dt} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \\
 a_z &= \frac{dw}{dt} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}
 \end{aligned}$$

Question

➤ Steady and Uniform Flow

- Steady Flow: Fluid characteristics at any point in fluid flow do not change with time
- Uniform Flow: Fluid characteristics at any time do not change with space



Velocity and Acceleration

$$a_x = \frac{Du}{Dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

$$a_y = \frac{Dv}{Dt} = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$$

$$a_z = \frac{Dw}{Dt} = \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z}$$

Material/substantial
derivative

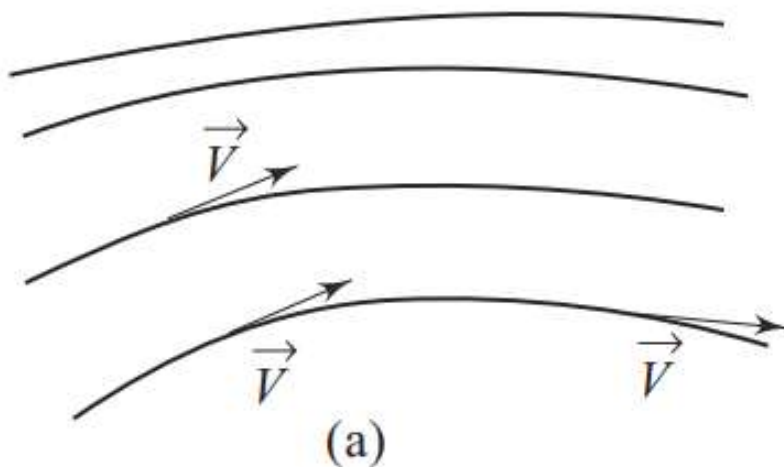
Convective derivative

$$\boxed{\frac{D}{Dt}} = \boxed{\frac{\partial}{\partial t}} + \boxed{u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}}$$

Local/temporal
derivative

Streamlines

- A streamline at **any time instant** can be defined as an imaginary curve in the flow field such that the tangent to the curve at any point represents the direction of instantaneous velocity at that point.
- Streamlines are defined for a particular time instant (Eulerian approach)
- Two streamlines cannot intersect



$$\vec{V} = \vec{i}u + \vec{j}v + \vec{k}w \quad \text{and} \quad d\vec{S} = \vec{i} dx + \vec{j}dy + \vec{k} dz$$

$$\vec{V} \times d\vec{S} = 0$$

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Path Lines

- Path line is the trajectory of a particular fluid particle (Lagrangian approach)
- Path lines inherently involves **variation of time**
- Two path lines can intersect, or one path line can itself form a loop

